

LLD Practice Platform

2-Day Engineering Assignment

The challenge

Design and build a small practice experience that helps a learner practice Low-Level Design, submit a solution, and receive useful, explainable feedback.

1. Background

LLD practice is often easy to start but difficult to evaluate. A learner can design a Parking Lot, Elevator, Vending Machine, or similar problem and still be unsure whether the responsibilities, abstractions, relationships, and trade-offs are actually good.

We want to explore a focused product that helps learners practice LLD repeatedly and understand where their design can improve.

2. Your Task

Research the problem briefly, define a focused MVP, and build a working prototype of an LLD practice platform. Your solution should concentrate on the learner journey rather than on building an LMS or a large assessment system.

A useful starting journey is:

Practice loop

Choose problem → Think / design → Submit → Get feedback → Review → Try again

3. What Your MVP Should Demonstrate

Area	Expected outcome
Problem	A small set of LLD problems with clear requirements and enough context to attempt them.
Practice	A learner can start an attempt and work on a solution in a form you choose (for example text, code, diagram, or a combination).
Submission	The learner can submit a solution and see its status.
Feedback	The platform provides useful feedback on the design. AI may be used where reasoning is helpful.
History	The learner can see previous attempts so the product supports improvement, not just one-time solving.
Core design	The important domain behaviour is represented with clear classes/interfaces and responsibilities.

4. The Main Design Questions

Your research and implementation should help answer these questions:

- What does a learner actually need to provide for an LLD practice attempt to be meaningful?
- What makes feedback useful when there can be more than one valid LLD solution?
- Which parts of evaluation should be deterministic, and which parts benefit from an LLM?
- How would your design accommodate another evaluation approach or another submission format later?
- What should happen if evaluation takes time or fails? Keep this practical; do not turn the assignment into a distributed-systems project.

5. Scope Boundary

This is primarily an LLD/domain-design exercise. Do not spend the majority of your time on Kubernetes, microservices, multi-region deployment, sharding, CDN design, or other large-scale HLD concerns. A simple monolith is completely acceptable.

Important distinction
LLD focus: classes, objects, responsibilities, interfaces, behaviour, relationships, patterns, extensibility, and code-level decisions. HLD is only a light consideration when explaining how your prototype would handle more users or slower AI evaluation.

6. AI Usage

You are encouraged to use ChatGPT, Claude, Gemini, Cursor, Copilot, or similar tools. We are evaluating how well you use AI together with your own judgement—not whether you avoid AI.

Include a short AI_USAGE.md explaining 3–5 meaningful AI-assisted decisions: what the AI suggested, what you accepted or rejected, and why.

7. What to Submit

Deliverable	What we expect
Research note	1–2 pages: learner problem, a few existing approaches/tools researched, key gaps, and your product direction.
Design note	A concise explanation of your MVP, user flow, important classes/interfaces, evaluation approach, and key trade-offs.
Working prototype	A demonstrable end-to-end practice flow from problem selection to feedback and attempt history.
Tests	Tests for important behaviour and at least a few failure/edge cases.
README + AI_USAGE.md	How to run the project, key decisions, limitations, and meaningful AI usage.

8. Time & Evaluation

Time: **2 days**. We expect a focused prototype, not a production-ready system.

Evaluation area	Weight
Problem understanding & research	15%
Product thinking / creativity	15%
LLD / domain design	25%
Evaluation & feedback approach	15%
Extensibility & engineering judgement	10%
Implementation quality	10%
Testing & reliability	5%
AI usage	5%

9. Submission

Please submit your completed requirements via this Google Form:

<https://docs.google.com/forms/d/e/1FAIpQLSfoHirkkjOeAYvzWksEVXSEQN5rEeD2Dyn5IPbSOkPhW65WHg/viewform?usp=publish-editor>.